

Monte Carlo Portfolio Simulation Tool

Product Requirements Document

"Through the Noise" — Full Build Specification

Version 1.0 | For Claude Code

1. Project Overview

Build a browser-based Monte Carlo retirement portfolio simulation tool for the "Through the Noise" YouTube channel. The tool is desktop-only, hosted as a static site on Render, and runs entirely client-side with no backend. All simulation logic executes in a Web Worker thread to prevent UI blocking.

The tool helps pre-retirees (ages 55–70) understand the probability that their investment portfolio will sustain their retirement spending across a range of market scenarios drawn from real historical data.

Data file: `simba_returns_data.json` — located in the project root directory. This file must be fetched at app startup via `fetch('./simba_returns_data.json')`. Do not hardcode return data in JavaScript.

2. Brand & Visual Identity

2.1 Color Palette (PLACEHOLDER — swap with brand kit values)

`--color-primary-dark: #1a1f2e /* Dark charcoal/slate background */`

`--color-primary-blue: #3b82f6 /* Electric blue — primary accent */`

`--color-primary-gold: #f59e0b /* Gold — achievement/highlight moments */`

`--color-surface: #252b3b /* Card/panel background */`

`--color-surface-light: #2e3547 /* Input field background */`

--color-text-primary: #f1f5f9 /* Primary text */

--color-text-secondary: #94a3b8 /* Muted/secondary text */

--color-text-tertiary: #64748b /* Hints, labels */

--color-border: #374151 /* Borders */

--color-success: #10b981 /* Positive outcomes */

--color-danger: #ef4444 /* Depleted portfolios / warnings */

--color-warning: #f59e0b /* Data quality warnings */

NOTE TO CLAUDE CODE: Replace all placeholder hex values above with the exact values from the "Through the Noise" brand kit before finalizing. The brand kit is in the local project folder.

2.2 Typography

- Primary font: System sans-serif stack (-apple-system, BlinkMacSystemFont, 'Segoe UI', sans-serif)
- Monospace for numbers in results tables: 'SF Mono', 'Fira Code', monospace
- **NOTE TO CLAUDE CODE:** Update to brand kit fonts if specified.

2.3 Page Header

- Title: **"Monte Carlo Portfolio Simulation"**
- Subtitle: **"Presented by Through the Noise"**
- Include Through the Noise logo (placeholder if file not found — display text logo)
- Tagline: *"Actionable tools to stop planning and start living"*
- Dark background header consistent with channel branding

2.4 Layout

Three-panel layout on desktop (min-width 1024px):

- **Left panel** (380px fixed width): All user inputs, stacked vertically with scroll
- **Center/right panel** (remaining width): Fan chart dominant, results below
- **Full width below panels:** Results summary text, metrics table, asset reference table, disclaimers

Display a "best viewed on desktop" banner for viewports under 768px. Do not attempt to make the tool fully responsive — desktop-only is acceptable.

3. Application Startup Sequence

1. Fetch `simba_returns_data.json` from project root
 2. Parse and validate JSON structure (check for `assets`, `annual_returns`, `metadata` keys)
 3. Display loading indicator while fetching
 4. On load success: render the full UI with default values pre-filled
 5. On load failure: display a clear error message explaining the data file could not be loaded
 6. Initialize Web Worker for simulation engine (see Section 9)
 7. Display the educational disclaimer checkbox (unchecked) — simulation cannot run until checked
-

4. Input Panel (Left Panel)

All inputs are in the left panel, stacked top to bottom in this order. Group inputs under clearly labeled section headers.

4.1 Section: "Your Portfolio"

4.1.1 Asset Allocation Table

- A table with up to 10 rows
- Each row contains:
 - **Asset Class dropdown:** All 22 asset classes from `simba_returns_data.json`, grouped by category (US Equity, International Equity, Fixed Income, Alternatives). Include a blank "— Select asset —" option.
 - **Allocation % input:** Numeric input, 0–100, integer only
 - **Remove row button:** × icon, removes the row (minimum 1 row must remain)
- **"Add Asset Class" button:** Adds a new blank row (disabled when 10 rows exist)
- **Allocation total indicator:** Live-updating display showing current total %.
 - Green when exactly 100%
 - Red with message "Allocations must sum to 100%" when not 100%
- **Default pre-fill on load:**
 - Row 1: S&P 500 — 60%
 - Row 2: Total Bond Market — 40%

4.1.2 Initial Portfolio Balance

- Label: "Starting Portfolio Balance"
- Input: Currency input, formatted with \$ and commas on blur
- Minimum: \$1,000
- Default: \$1,000,000
- Note below field: "Enter your total investable portfolio value at retirement start"

4.2 Section: "Simulation Settings"

4.2.1 Current Age

- Label: "Your Current Age"
- Input: Integer dropdown, 40–80
- Default: 60

4.2.2 Simulation Period

- Label: "Simulation Length"
- Input: Dropdown, options: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50 years
- Default: 30 years
- Note: "Full calendar years are simulated (e.g. 30 years = 30 annual periods)"
- When changed: dynamically update the Expense Schedule section (Section 4.3)

4.2.3 Number of Simulations

- Label: "Number of Simulations"
- Input: Dropdown
 - 1,000 simulations (faster)
 - 5,000 simulations
 - 10,000 simulations (recommended)
- Default: 10,000
- Note below: "More simulations = greater accuracy, longer run time"

4.2.4 Historical Data Period

- Label: "Historical Return Period"
- Input: Dropdown with these options:
 - "Full history — Native data per asset (1871–2025)"
 - "Post-WWII — 1946 to 2025"
 - "Modern era — 1976 to 2025 (recommended)"
 - "Custom range — Select start and end year"
- Default: "Modern era — 1976 to 2025"
- When "Custom range" is selected: show two additional year dropdowns
 - Start year: 1871 to 2020 (integer, dropdown or input)

- End year: start year + 5 to 2025
- **Data quality warning logic:** When any selected asset class has a native start date AFTER the selected period start year, display a specific warning:
 - Warning format: "⚠ [Asset Name] has native data starting in [year]. Using [selected period start] will include reconstructed proxy data prior to [native start year], which may affect simulation accuracy."
 - Show one warning per constrained asset (only the most constrained asset if multiple are affected — the one with the latest native start date)
 - Warning appears directly below the Historical Data Period dropdown
 - Warning uses gold/amber color consistent with brand

4.2.5 Sequence of Returns Risk

- Label: "Sequence of Returns Stress Test"
- Input: Toggle switch (on/off)
- Default: Off
- When ON: display note "The actual 2008 annual returns will be applied to Year 1 of every simulation. This stress-tests your portfolio against the worst recent market conditions."
- When ON: 2008 returns from `simba_returns_data.json` are injected as year 1 of every simulation path. Subsequent years continue with normal bootstrap sampling.

4.2.6 Inflation Adjustment

- Label: "Inflation-Adjust Expenses"
- Input: Toggle switch (on/off)
- Default: On
- When ON: expenses entered in today's dollars are inflated annually using the bootstrap year's historical inflation draw from `simba_returns_data.json`
- When OFF: expenses are treated as fixed nominal amounts — display warning "Turning off inflation adjustment means your real purchasing power will decline over time. This is not recommended for most users."
- Note always visible: "Enter all expenses in today's dollars. The model will apply historical inflation annually."

4.3 Section: "Expense & Income Schedule"

This section is dynamically generated based on the Simulation Period selected (Section 4.2.2).

4.3.1 Expense Period Buckets

- Number of buckets = Simulation Period ÷ 5 (e.g. 30 years = 6 buckets)
- Each bucket label shows the age range:
 - Bucket 1: "Ages [current age] to [current age + 4] (Years 1–5)"
 - Bucket 2: "Ages [current age + 5] to [current age + 9] (Years 6–10)"

- And so on...
- Each bucket contains the following fields:

Annual Expenses

- Label: "Annual Expenses (today's dollars)"
- Input: Currency input, formatted with \$ and commas
- Default: blank on first bucket; subsequent buckets auto-fill from the previous bucket value on change (carry-forward behavior)
- Carry-forward indicator: Show a small "↓ carried forward" label in muted text next to any bucket that has been auto-filled. When the user manually edits a carried-forward value, the indicator disappears.
- Note once above entire expense section: "Enter expenses in today's dollars. The model will inflation-adjust these amounts annually. All amounts are pre-tax — account for income taxes within your expense figures."

Expense Entry Mode (shown once above all buckets, applies to all)

- Label: "Expense entry mode"
- Toggle: "Annual" / "Monthly"
- Default: Annual
- When Monthly selected: all expense inputs show monthly amounts; model multiplies by 12 internally
- Note: "Switch between annual and monthly entry. The model always calculates annually."

Social Security Income

- Label: "Social Security Income"
- Input: Currency input (annual amount in today's dollars)
- Sub-label: "Start Age" — integer input, 62–70, default 67
- Default value: \$0
- Carry-forward behavior: same as expenses — auto-fills from previous bucket; shows "↓ carried forward" indicator
- SS income is applied starting in the exact year the user reaches the SS Start Age, not at the bucket boundary
- SS income receives annual COLA equal to the bootstrap year's inflation draw (same inflation rate applied to expenses)
- Since expenses are in today's dollars and SS is entered in today's dollars, both are inflated forward together — the model does not need to discount SS separately

Pension Income

- Label: "Pension Income"
- Input: Currency input (annual amount in today's dollars)
- Sub-label: "Start Age" — integer input, 50–80, default 65
- Sub-label: "Includes COLA?" — Toggle Yes/No, default No
 - If Yes: pension receives the same annual COLA as SS (bootstrap year inflation)
 - If No: pension amount is fixed in nominal dollars (not inflation-adjusted)
- Default value: \$0
- Carry-forward behavior: same as expenses

Net Portfolio Withdrawal (read-only, calculated)

- Label: "Est. Net Portfolio Draw"
- Display: Calculated field showing Expenses - SS - Pension for that bucket (in today's dollars)
- Color: Green if positive (portfolio needed), blue if zero, amber if negative (income exceeds expenses — surplus)
- If negative: show note "Income exceeds expenses in this period. Surplus will be added to portfolio."
- This is a display-only field to help users understand the net portfolio impact

4.3.2 Annual Rebalancing Note

- Display as a static info box (not user-adjustable):
- Text: "i Your portfolio will be rebalanced to your target allocation annually, after each year's withdrawal is taken."

4.4 Section: "Run Simulation"

4.4.1 Educational Disclaimer Checkbox

- Checkbox (unchecked by default, required before each simulation run)
- Label: "I understand this tool is for educational purposes only and does not constitute financial, investment, tax, or legal advice. Results are hypothetical and past performance does not guarantee future results."
- The "Run Simulation" button is disabled until this checkbox is checked
- Checkbox resets to unchecked after each simulation completes (user must re-acknowledge before running again)

4.4.2 Run Simulation Button

- Large, prominent button: "Run Simulation"
- Disabled states:
 - Allocations do not sum to 100%
 - Disclaimer checkbox unchecked

- No expense entered in bucket 1
- Simulation currently running
- Active state: Electric blue background, white text
- On click: validates all inputs, then posts message to Web Worker

4.4.3 Reset Button

- Smaller secondary button: "Reset to Defaults"
- Resets all inputs to their default values as specified in this document
- Does not reset the results panel (results remain visible until a new simulation runs or reset is confirmed)
- Requires confirmation: "Are you sure you want to reset all inputs?"

5. Simulation Engine (Web Worker)

CRITICAL: The entire simulation must run in a Web Worker (`simulation.worker.js`). Never run simulation loops on the main thread.

5.1 Worker Communication

- Main thread posts inputs object to worker via `worker.postMessage(inputs)`
- Worker posts progress updates back: `{ type: 'progress', completed: N, total: N }`
- Worker posts final results: `{ type: 'results', data: resultsObject }`
- Worker posts errors: `{ type: 'error', message: string }`

5.2 Progress Indicator

- While simulation runs: display animated counter in results panel
- Format: "Simulating... 4,832 / 10,000"
- Update frequency: every 100 completed simulations
- Show a subtle progress bar below the counter

5.3 Bootstrap Sampling Logic

Year Package Selection

1. Determine the eligible year pool: all years in `annual_returns` array where:
 - Year falls within the selected historical period (start year to end year)
 - All selected asset classes have non-null return values for that year
2. On each simulation, for each year of the simulation period:
 - Randomly select one year from the eligible pool (uniform random, with replacement)

- Apply ALL return values from that year's record simultaneously (preserves cross-asset correlations)
 - Apply the inflation value from that SAME year's record (preserves inflation-return correlation)
3. **Sequence of Returns stress test** (if enabled):
- Force year 1 of every simulation to use the 2008 data record from `annual_returns`
 - Years 2 through N continue with normal random bootstrap sampling

Annual Portfolio Calculation

For each year t in each simulation:

1. Determine expense for year t:

- Look up which 5-year bucket year t falls in
- Get the bucket's expense amount (today's dollars)
- If inflation adjustment ON: apply cumulative inflation from start to year t
- If inflation adjustment OFF: use nominal amount as-is
- Convert to annual if user entered monthly (multiply by 12)

2. Determine guaranteed income for year t:

- Social Security: if `current_age + t` $\geq$ `ss_start_age`, apply SS amount
- Inflate SS by cumulative inflation from start to year t (COLA)
- Otherwise: 0
- Pension: if `current_age + t` $\geq$ `pension_start_age`, apply pension amount
- If pension has COLA: inflate by cumulative inflation
- If no COLA: use nominal pension amount (no adjustment)
- Otherwise: 0

3. Net withdrawal = $\text{MAX}(0, \text{inflated_expense} - \text{ss_income} - \text{pension_income})$

- If income > expense: surplus = income - expense, add surplus to portfolio

4. Post-withdrawal balance = portfolio_balance - net_withdrawal

- If post-withdrawal balance <= 0: portfolio is depleted
- Record depletion year = t
- Remaining years in this simulation = \$0
- Break inner loop

5. Apply portfolio return for year t:

- Calculate weighted portfolio return from bootstrap year:

weighted_return = sum(allocation_pct[i] * asset_return[i] / 100) for all assets i

- post_return_balance = post_withdrawal_balance * (1 + weighted_return / 100)

6. Rebalance: allocation percentages are maintained — no action needed in calculation

(rebalancing is implicit — we always apply the same target allocation weights)

7. Record year t balance for this simulation path

Inflation Tracking

- Maintain cumulative inflation index for each simulation path
- Year 0: inflation_index = 1.0
- Each year: inflation_index *= (1 + bootstrap_year_inflation / 100)
- Real portfolio balance = nominal_balance / inflation_index

5.4 Results Object Structure

After all simulations complete, the worker computes and returns:

```
{  
  
  inputs_summary: {  
  
    n_simulations: 10000,  
  
    period_years: 30,
```

```

start_age: 60,

end_age: 90,

initial_balance: 1000000,

historical_period: "1976-2025",

portfolio_historical_cagr: 10.10, // from selected period stats

portfolio_historical_mean: 10.50,

portfolio_historical_std: 10.09,

inflation_historical_mean: 3.91,

inflation_historical_std: 1.30,

constraining_asset: "TIPS", // asset with latest native start

constraining_asset_start: 1985,

sequence_of_returns_active: false,

rebalancing: "annual_post_withdrawal"

},

// All simulation paths (for fan chart) — array of N_SIM arrays of length YEARS

// Store at 10th/25th/50th/75th/90th percentile paths only to limit memory

// Each path is array of annual nominal balances

percentile_paths: {

  p10: [...], // array of length period_years+1 (including year 0 = initial balance)

  p25: [...],

  p50: [...],

```

```

p75: [...],

p90: [...],

real_p10: [...], // inflation-adjusted versions

real_p25: [...],

real_p50: [...],

real_p75: [...],

real_p90: [...],

},

// Summary statistics at each percentile

statistics: {

  p10: { ending_balance_nominal, ending_balance_real, cagr_nominal, cagr_real,

        annualized_volatility, sharpe_ratio, max_drawdown_pct, depleted: bool,

        depletion_year: null },

  p25: { ... },

  p50: { ... },

  p75: { ... },

  p90: { ... },

},

// Portfolio success metrics

success_metrics: {

  success_count: 7432, // simulations ending with balance > 0

  total_simulations: 10000,

```

```

    success_rate_pct: 74.32,    // success_count / total * 100

    failure_count: 2568,

    avg_depletion_year: 22.4,    // average year of depletion among failed sims

    median_depletion_year: 24,    // median year of depletion among failed sims

    pct_depleted_by_year: [...], // array: % of all sims depleted by each year

},

// Raw data for results table (one row per percentile)

results_table: [

    { percentile: 10, label: "10th percentile (pessimistic)",

      ending_nominal, ending_real, cagr_nominal, cagr_real,

      volatility, sharpe, max_drawdown, depleted, depletion_year },

    { percentile: 25, label: "25th percentile" ... },

    { percentile: 50, label: "50th percentile (median)" ... },

    { percentile: 75, label: "75th percentile" ... },

    { percentile: 90, label: "90th percentile (optimistic)" ... },

]

}

```

5.5 Metric Definitions for Simulation Results

CAGR (nominal): $(\text{ending_balance} / \text{initial_balance}) ^ { (1 / \text{years}) } - 1$,
expressed as %

CAGR (real): $(\text{ending_balance_real} / \text{initial_balance}) ^ { (1 / \text{years}) } - 1$,
expressed as %

Annualized Volatility: Standard deviation of annual portfolio returns across all years in that simulation path, expressed as %

Sharpe Ratio: $(\text{mean_annual_return} - \text{mean_annual_tbill_return}) / \text{std_dev_annual_return}$

- Use the T-Bill returns from the same bootstrap years drawn in that simulation
- Note in UI: "Sharpe ratio uses arithmetic mean return minus average T-Bill rate"

Maximum Drawdown: Largest peak-to-trough decline in portfolio value as a percentage, inclusive of withdrawals.

- Formula: for each year t , track $\text{running_peak} = \max(\text{balance}[0..t])$
- Drawdown at $t = (\text{running_peak} - \text{balance}[t]) / \text{running_peak}$
- Max drawdown = $\max(\text{drawdown}[t])$ across all years t
- Expressed as negative percentage (e.g. -34.2%)

Depletion Year: The simulation year (1-indexed) in which the portfolio first reaches \$0 or below. Null if portfolio survives full period.

6. Results Display (Center/Right Panel + Full Width)

6.1 Fan Chart (Center Panel, Dominant Visual)

The fan chart is the primary output visual and should occupy the largest portion of the results area.

Chart library: Use Chart.js (already in CDN allowlist at cdnjs.cloudflare.com)

Chart type: Line chart with filled areas between percentile bands

X-axis:

- Labels: Ages (current_age to $\text{current_age} + \text{period_years}$)
- Show every 5 years as labeled tick

Y-axis:

- Label: "Portfolio Balance"
- Format: Abbreviated currency (\$500K, \$1M, \$1.5M)

- Start at \$0

Toggle: "Nominal" / "Real (inflation-adjusted)" — default Nominal

- Toggle updates chart data and y-axis label without re-running simulation

Percentile bands (from bottom to top):

- Outer band (10th–90th): lightest blue fill, very low opacity (0.15)
- Inner band (25th–75th): medium blue fill, low opacity (0.25)
- Median line (50th): solid electric blue line, 2.5px width
- No fill on median line itself

Additional elements:

- Horizontal dashed line at \$0 (red, 1px, labeled "Portfolio depleted")
- Starting balance dot at year 0 (gold dot on median line)
- Gold accent annotation showing the Portfolio Success Rate: e.g. "74.3% Success Rate" placed in upper right of chart area
- Legend below chart: colored swatches for each band + median

Depleted scenario shading:

- If any simulations depleted, show a shaded red region below \$0 line
- Small annotation: "[N] of [total] simulations depleted before year [end]"

Chart title: "Projected Portfolio Balance — [N] Simulations"

6.2 Headline Success Metric

Displayed prominently above or beside the fan chart:

Large display showing:

Portfolio Success Rate

74.3%

7,432 of 10,000 simulations ended with \$1 or more

Color coding:

- $\geq 90\%$: Green

- 75–89%: Electric blue
- 50–74%: Amber/gold
- < 50%: Red

Below success rate, show:

Among simulations that depleted:

Average depletion: Year 22 (Age 82)

Median depletion: Year 24 (Age 84)

6.3 Results Summary Text Block

Displayed below the fan chart, full width. Dynamically generated to match the example format provided:

Monte Carlo simulation results for [N] portfolios with \$[initial_balance] initial portfolio balance using available historical returns data from Jan [start_year] to Dec [end_year] with annual sampling. The historical pre-tax return for the selected portfolio for this period was [mean]% mean return ([cagr]% CAGR) with [std]% standard deviation of annual returns.

The simulation results are based on generated nominal returns and specified inflation-adjusted withdrawals ([expense] per year in today's dollars, Bucket 1). The simulated inflation model used historical inflation with [infl_mean]% mean and [infl_std]% standard deviation based on the Consumer Price Index (CPI-U) data from Jan [start_year] to Dec [end_year]. The generated inflation samples were correlated with simulated asset returns based on historical correlations.

[If constraining asset exists]: The available historical data for the simulation inputs was constrained by [Asset Name] ([ticker]) which has native data beginning [month] [year].

[If Sequence of Returns active]: This simulation applies 2008 historical returns to Year 1 of every portfolio path as a sequence-of-returns stress test.

All returns are pre-tax. Users should account for income taxes within their expense inputs.

Portfolio is rebalanced annually after each year's withdrawal.

All bracketed values are dynamically substituted from the results object.

6.4 Results Metrics Table

A table with one column per percentile (10th, 25th, 50th, 75th, 90th) and these rows:

Metric	10th	25th	50th	75th	90th
Ending Balance (nominal)					
Ending Balance (real)					
CAGR — nominal					
CAGR — real					
Annualized Volatility					
Sharpe Ratio					
Maximum Drawdown					
Portfolio Depleted?					
Depletion Year (if depleted)					

Formatting:

- Currency values: \$X,XXX,XXX format
- Percentages: X.XX%
- Sharpe: X.XXX

- Depleted: "Yes (Year N)" or "No"
- Column headers: "10th\nPessimistic", "25th", "50th\nMedian", "75th", "90th\nOptimistic"
- Highlight the 50th percentile column with a subtle background accent
- Color the "Ending Balance" rows: green if > initial_balance, amber if > 0 but < initial_balance, red if 0

Table note below: "CAGR and volatility reflect simulated portfolio paths. Sharpe ratio uses arithmetic mean return minus average T-Bill rate, divided by standard deviation. Maximum drawdown is the largest peak-to-trough portfolio value decline including withdrawals. Returns are pre-tax."

7. Asset Class Reference Table

Displayed below the results metrics table as a collapsible section.

Section header: "Asset Class Reference Data" with expand/collapse toggle — collapsed by default

When expanded, renders the full interactive asset class reference table:

- Identical to the standalone reference table built previously
 - Three period toggle buttons: Native / Post-WWII (1946–2025) / Modern era (1976–2025)
 - Columns: Asset Class, Ticker, First Year, Years, CAGR, Std Dev, Volatility bar, Sharpe, Worst Year, Best Year, Avg RF, Data Quality badge
 - Column headers are sortable (click to sort ascending/descending)
 - Summary strip at top: Highest CAGR, Best Sharpe, Avg US Equity CAGR, Avg RF Rate
 - Data quality badges: Native 1871 (green), Spliced <1927 (amber), Spliced <1970+ (red), Limited history (blue)
 - Data quality warning banner for non-modern periods
 - All data sourced from `simba_returns_data.json` assets metadata
-

8. PDF Export

8.1 Trigger

- "Download PDF Report" button displayed in results area after simulation completes
- Library: `jsPDF` + `html2canvas` (load from CDN)
- Button is hidden until simulation results are available

8.2 PDF Contents (in order)

1. **Header:** Through the Noise logo + "Monte Carlo Portfolio Simulation" title + date generated
2. **Input Summary:** Table showing all user inputs (allocation, initial balance, period, expenses per bucket, SS, pension, simulation settings)
3. **Headline metric:** Portfolio Success Rate (large, styled)
4. **Fan chart:** Captured via html2canvas as image
5. **Results summary text block:** Full dynamic text (Section 6.3)
6. **Results metrics table:** All 5 percentile columns (Section 6.4)
7. **Full legal disclaimer:** Complete text (Section 11)
8. **Footer:** "Generated by Through the Noise Monte Carlo Portfolio Simulation Tool | throughthe noise.com | For educational purposes only"

8.3 PDF Styling

- Dark branding on header/footer matching website
 - White background for body content (better print legibility)
 - Through the Noise branding consistent with brand kit
 - Page size: Letter (8.5" x 11")
 - Margins: 0.75" all sides
-

9. Web Worker Architecture

9.1 File Structure

project-root/

├── index.html

├── app.js (main thread — UI, inputs, results rendering)

├── simulation.worker.js (Web Worker — all simulation math)

├── simba_returns_data.json (data file)

├── styles.css (all styles)

├── render.yaml (Render deployment config)

└─ README.md (methodology documentation)

9.2 Worker Initialization

// In app.js

```
const worker = new Worker('./simulation.worker.js');

worker.onmessage = (e) => {

  if (e.data.type === 'progress') updateProgressDisplay(e.data);

  if (e.data.type === 'results') displayResults(e.data.data);

  if (e.data.type === 'error') displayError(e.data.message);

};

worker.onerror = (e) => displayError('Simulation worker error: ' + e.message);
```

9.3 Worker Must Not

- Import any external libraries (keep it pure JS for performance)
- Access the DOM
- Use `fetch` — the main thread passes the data in the inputs message

9.4 Performance Requirements

- 10,000 simulations × 30 years must complete in under 10 seconds on a modern laptop
- Progress updates every 100 simulations
- Results object size must be manageable — store only percentile paths, not all 10,000 raw paths

10. Render Deployment Configuration

Create `render.yaml` in project root:

services:

- type: web

name: monte-carlo-sim

env: static

buildCommand: ""

staticPublishPath: .

headers:

- path: /*

name: Cache-Control

value: no-cache

routes:

- type: rewrite

source: /*

destination: /index.html

Also create [README.md](#) documenting:

- Data source attribution (Simba Bogleheads Backtesting Spreadsheet)
- Bootstrap methodology explanation
- Null convention for pre-native-start years
- Three historical period definitions
- Annual rebalancing post-withdrawal convention
- Inflation correlation with asset returns
- Sharpe ratio calculation methodology

11. Legal Disclaimers

11.1 Inline Tax Warning (near expense inputs)

Display as a prominent amber/gold info box directly above the expense input section: "⚠ TAX NOTICE: All returns in this simulation are pre-tax. Your actual after-tax performance will depend on your individual tax situation, account types (IRA, 401k, taxable), and applicable tax rates."

Account for income taxes — including potential RMDs and capital gains — within your expense figures. Consult a qualified tax professional."

11.2 Disclaimer Checkbox (Section 4.4.1)

Required before every simulation run. Text: "I understand this tool is for educational purposes only and does not constitute financial, investment, tax, or legal advice. Results are hypothetical and past performance does not guarantee future results."

11.3 Full Legal Disclaimer (Page Footer)

Display as a collapsible section at the very bottom of the page labeled "Important Disclosures and Legal Information". Collapsed by default. Full text:

IMPORTANT DISCLOSURES AND LEGAL INFORMATION

Not Financial Advice. This tool is provided for educational and informational purposes only and does not constitute financial advice, investment advice, tax advice, or legal advice. Nothing in this tool should be construed as a recommendation to buy, sell, or hold any security or investment product, or to adopt any particular investment strategy.

Hypothetical Results. All simulation results are hypothetical in nature. They are based on historical data and mathematical modeling and do not represent actual investment outcomes, actual account performance, or the results any investor actually achieved. Hypothetical performance results have many inherent limitations and no representation is made that any account will or is likely to achieve results similar to those shown.

Past Performance. Past performance is not indicative of future results. Historical returns used in this simulation are not a guarantee, projection, or prediction of future performance. Actual investment returns will vary and may be negative.

No Warranty. This tool is provided "as is" without warranty of any kind, either express or implied, including but not limited to warranties of merchantability, fitness for a particular purpose, or non-infringement. Through the Noise makes no representations or warranties about the completeness, accuracy, reliability, suitability, or availability of the tool or the information it contains. Any reliance you place on such information is strictly at your own risk.

Data Sources. Historical return data is sourced from the Simba Bogleheads Backtesting Spreadsheet, a community-maintained open-source dataset. Some asset class data prior to fund inception dates has been reconstructed using proxy indices and statistical splicing methodology. Through the Noise does not warrant the accuracy or completeness of this

third-party data. The Simba Bogleheads dataset is used with appreciation for the Bogleheads community.

Methodology Limitations. Monte Carlo simulations are based on historical bootstrap sampling and represent one of many possible simulation methodologies. Different methodologies may produce materially different results. This simulation assumes annual rebalancing to target allocations, which may not reflect your actual investment behavior, transaction costs, tax consequences of rebalancing, or real-world constraints. The simulation does not account for investment management fees, brokerage commissions, or other transaction costs, which would reduce actual returns.

Tax Treatment. All returns shown are pre-tax. The actual after-tax performance of any investment will depend on your individual tax situation, account types, applicable tax rates, and potential required minimum distributions (RMDs). This tool does not account for taxes of any kind. Consult a qualified tax professional before making any investment or withdrawal decisions.

Social Security and Pension. Social Security and pension income projections entered by the user are assumed to be accurate. This tool does not independently verify benefit estimates. Actual benefits may differ based on future changes in government policy, employer plan terms, your earnings history, or other factors outside your control. Social Security benefit estimates should be obtained from the Social Security Administration at ssa.gov.

Inflation. Inflation assumptions are derived from historical CPI-U data. Future inflation may differ materially from historical patterns. This tool's inflation modeling is for illustrative purposes only.

Professional Advice. The results of this simulation should not be used as the sole or primary basis for any financial, investment, retirement, tax, or legal decision. You are strongly encouraged to consult with a qualified financial advisor, Certified Financial Planner (CFP), tax professional, or legal counsel before making any financial decisions. A qualified advisor can account for your complete financial picture, including factors not captured by this tool.

Limitation of Liability. In no event shall Through the Noise, its creators, contributors, or affiliates be liable for any direct, indirect, incidental, special, consequential, or punitive damages arising out of or in connection with your use of this tool or reliance on its results.

This tool is presented by Through the Noise for educational purposes. Through the Noise is not a registered investment advisor, broker-dealer, or financial planning firm.

11.4 Disclaimer in PDF Export

The full disclaimer text (Section 11.3) must appear in its entirety on the final page(s) of every PDF export.

12. Input Validation Rules

Input	Rule
Asset allocations	Must sum to exactly 100% before simulation can run
Initial balance	Minimum \$1,000, maximum \$99,999,999
At least one asset	Must have at least 1 asset class selected
Expense Bucket 1	Must have a value > 0
Age	Must be between 40 and 80
SS start age	Must be between 62 and 70
Pension start age	Must be between 50 and 80
Custom year range	End year must be at least 5 years after start year
Disclaimer checkbox	Must be checked before each run
Duplicate asset classes	Do not allow the same asset class to be selected in two rows

13. Error States

Scenario	User-facing message
Data file fails to load	"Unable to load historical return data. Please check that simba_returns_data.json is in the project root and refresh the page."

Scenario	User-facing message
Web Worker not supported	"Your browser does not support Web Workers. Please use a modern browser (Chrome, Firefox, Safari, Edge)."
No eligible years in pool	"No historical years match your selected period and asset classes. Please select a different historical period or asset combination."
Simulation worker crash	"The simulation encountered an unexpected error. Please reset and try again. If the problem persists, try reducing the number of simulations."
PDF generation failure	"PDF export failed. Please try again or use your browser's print function (Ctrl+P / Cmd+P) as an alternative."

14. Build Sequence Recommendation for Claude Code

Build in this order to avoid architectural rework:

Phase 1 — Data layer

- Fetch and parse `simba_returns_data.json`
- Build the asset class reference table (Section 7) — this validates data is loading correctly
- Verify all 22 assets, 155 years, statistics display correctly

Phase 2 — Simulation engine

- Build `simulation.worker.js` with bootstrap logic
- Test with hardcoded inputs before connecting to UI
- Verify results object structure matches Section 5.4
- Validate: crisis years (1931, 2008) appear in simulations, inflation correlates with returns

Phase 3 — Input panel

- Build all input controls (Sections 4.1–4.4)
- Connect to simulation worker

- Implement all validation rules (Section 12)
- Test dynamic expense buckets with different simulation periods

Phase 4 — Results display

- Fan chart (Section 6.1) — connect to worker results
- Headline success metric (Section 6.2)
- Results summary text block (Section 6.3)
- Results metrics table (Section 6.4)

Phase 5 — Polish and export

- PDF export (Section 8)
- Reset functionality
- Progress indicator
- All disclaimers and legal text (Section 11)
- Render deployment config (Section 10)
- Final brand kit color/font substitution

15. Technical Constraints and Notes

- **No backend required.** Pure static HTML/CSS/JS.
 - **No npm build step required** unless Claude Code prefers it. Vanilla JS is acceptable and preferred for simplicity.
 - **CDN libraries allowed:** Chart.js, jsPDF, html2canvas (all available on cdnjs.cloudflare.com)
 - **No React, Vue, or other frameworks** — vanilla JS only for maximum compatibility and simplicity on Render static hosting
 - **Browser target:** Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
 - **Minimum viewport:** 1024px wide — display "best viewed on desktop" message below this
 - **Data file size:** `simba_returns_data.json` is approximately 131KB — acceptable for direct fetch on page load
 - **No localStorage or sessionStorage** — no data persistence between sessions
 - **All monetary calculations** use floating point — round to 2 decimal places for display, full precision internally
 - **Random number generation:** Use `Math.random()` — cryptographic randomness is not required for Monte Carlo simulation
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End of Requirements Document Version 1.0 — Through the Noise Monte Carlo Portfolio Simulation